Does More Context Help? Evaluating Knowledge-Augmented Prompt Strategies for Arabic Post-OCR Correction with a Lightweight LLM

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***Abstract*—Arabic optical character recognition (OCR) systems produce systematic character-level errors—dot confusions, hamza substitutions, taa marbuta swaps—that degrade downstream text processing. Post-OCR correction using large language models (LLMs) is a natural remedy, but it is unclear how much additional context actually helps, and whether findings generalize across font families, script types, and model scales. This paper reports a three-experiment controlled study using Qwen3-4B-Instruct-2507 as the primary correction model. Experiment 1 evaluates eight prompt-engineering strategies across nine datasets (eight typewritten fonts from PATS-A01 and KHATT handwritten). Experiment 2 extends the evaluation to thirteen datasets, adding KHATT-Paragraph (handwritten paragraphs), Yarmouk (contemporary printed documents), Muharaf (handwritten manuscripts), and Historical (archival pages), testing eight prompt configurations. Experiment 3 benchmarks four LLMs across two OCR sources and two held-out benchmarks (RDI-Test-Lines, Kitab). All metrics are computed on normal samples only (OCR/GT length ratio ≤ 5) with diacritics stripped before CER/WER computation.**

In Experiment 1, we evaluate eight prompt-engineering strategies. The central finding is that zero-shot LLM correction *harms* the PATS-A01 macro-average CER (5.57% $\rightarrow$ 5.84%, +4.8%), degrading 7 of 8 fonts. Only domain-matched BM25 RAG (Phase 8) beats the OCR baseline (5.45%, -0.12 pp). On KHATT (34.24% CER), all strategies improve by 0.4–1.1 pp; the best is Error-Signature RAG (33.13%, -1.11 pp).

In Experiment 2, conservative zero-shot prompting (T3) achieves the best PATS result: 5.27% CER (-5.4% relative), outperforming RAG without retrieval overhead. Base+RAG (T2) is the best overall configuration (PATS 5.44%, KHATT 33.82%) and is used in Experiment 3.

In Experiment 3, four models are compared with T2 across multiple domains. Gemma-3-12B achieves 5.33% PATS CER; Qwen3-4B is best on KHATT (33.82%) and full-page datasets (OCR 86.50% $\rightarrow$ corrected 68.18%). LLM correction generalises to unseen benchmarks: RDI-Test-Lines -10.7%, Kitab Benchmark $-17 \rightarrow -24$% CER reduction. These findings establish that over-correction is universal at low OCR quality, that conservative prompting is more effective than knowledge augmentation for clean typewritten text, and that RAG-anchored correction generalises to diverse unseen domains.

***Index Terms*—Arabic OCR, post-OCR correction, large language models, prompt engineering, retrieval-augmented generation, BM25, Qwen3, PATS-A01, KHATT, KHATT-Paragraph, Yarmouk, Muharaf, historical manuscripts, multi-domain evaluation**

I. Introduction

Arabic optical character recognition occupies a uniquely difficult position in the document analysis landscape. The script is inherently cursive, with letter forms that vary by position within the word (isolated, initial, medial, final); dots serve as the sole phonemic and semantic distinguisher for entire character groups (*ba*/*ta*/*tha*/*ya*/*nun* share an identical skeletal form); and the optional nature of diacritics introduces a pervasive ambiguity that complicates both recognition and evaluation. These properties conspire to produce systematic, structured errors in open-source OCR engines such as Qaari [1]—errors that propagate unchecked into downstream pipelines for information retrieval, machine translation, and text-to-speech synthesis.

Post-OCR correction—treating recognition output as noisy text and applying a language model as a denoiser—has attracted renewed attention since instruction-tuned large language models (LLMs) became widely available. The paradigm is appealing: it requires no task-specific training data, operates in a single forward pass, and leverages the model's implicit prior over valid linguistic sequences. Yet two questions remain inadequately addressed in the literature. First, *does injecting structured knowledge about the OCR engine's failure modes into the prompt yield measurable gains over zero-shot correction?* Second, and more fundamentally, *does correction effectiveness generalise across font families and script types, or is it an artefact of the specific evaluation conditions?*

To answer these questions, we design a controlled multi-dataset experiment that starts from a zero-shot baseline and progressively augments the prompt with orthogonal knowledge sources (a failure-filtered confusion matrix, self-reflective diagnostics, morphological post-processing, and retrieval), each evaluated in isolation against the zero-shot baseline.

The central finding is that correction effectiveness is governed by a *font-dependent over-correction threshold*: on moderate-to-high-error fonts (10–25% CER) zero-shot correction yields large reductions, but on the lowest-error font and on segmentation-dominated handwritten text it is net harmful or marginal.

The contributions of this work are:

- A three-experiment systematic study across thirteen datasets spanning CER 2.12–129%, revealing font- and domain-dependent behaviour that single-dataset studies structurally cannot detect.
- A formal characterisation of the *over-correction threshold*: the baseline CER below which LLM correction is net harmful, estimated at ≈6–7% for Qwen3-4B on Qaari output.
- A novel self-reflective prompting strategy that injects the model's own training-split error statistics and word-level failure pairs into the system prompt, giving the strongest isolated PATS gain (−0.7 pp CER) and uniquely improving the lowest-error font.
- A domain-matched BM25 RAG strategy that retrieves similar correction pairs as per-sample few-shot context; domain alignment avoids the catastrophic degradation of external-corpus RAG and is the only Experiment 1 strategy to beat the OCR baseline on PATS-A01.

II. Related Work

Arabic OCR has been studied for over three decades [2], and its core difficulties are structural: the cursive baseline couples segmentation with recognition, diacritics are contextually optional yet semantically significant, and several letter groups (ba/ta/tha/ya/nun; fa/qaf; dal/dhal; ra/zay) share identical skeletal forms differing only in dots. These dot-group confusions are the dominant error type in modern engines and the primary target of the confusion-matrix augmentation evaluated here. Qaari, the open-source engine used throughout, is a vision–language model (VLM) fine-tuned from Qwen2-VL-2B [1]; it exhibits systematic, font-dependent dot-confusion patterns that are quantifiable.

Post-OCR correction has evolved through three paradigms. Rule-based and statistical methods used lexicon lookup, n-gram models, and noisy-channel formulations [7]; they are effective for frequent patterns but require manual curation and struggle with unseen errors. Neural sequence-to-sequence approaches treat correction as translation over parallel (OCR, GT) corpora [8], [9], but need substantial training data and are tightly coupled to a specific engine's error distribution. LLM-based methods—the paradigm explored here—correct OCR output via natural-language prompts without task-specific training and can outperform specialised models on English [10], yet Arabic evaluations remain limited to single-dataset BERT-class studies [11]. No prior work systematically evaluates multiple prompt augmentations across multiple Arabic fonts under a controlled isolation design.

Knowledge injection into prompts yields consistent NLP gains: few-shot exemplars activate task priors [12], chain-of-thought decomposes reasoning [13], and retrieval-augmented generation grounds outputs in external evidence [14]. Most relevant, Reflexion [15] feeds an agent its own prior failures as verbal reinforcement; our Phase 4 self-reflective strategy adapts this to a single-turn correction task, giving the model a statistical summary of its training-split errors and concrete word-level failure pairs—to our knowledge the first application of self-diagnostic prompting to post-OCR correction in any language. We further use CAMeL Tools [5], [16] for morphological validation (Phase 5) and error characterisation (Phase 1), and evaluate on the complementary PATS-A01 [3] (eight fonts over identical text, enabling controlled cross-font comparison) and KHATT [4] (real handwritten, segmentation-dominated, far higher error rates) benchmarks.

TABLE I

Evaluation datasets. Exp1 = Experiment 1 (9 datasets). Exp2 adds four full-page datasets (total 13). OCR CER = Qaari baseline (normal samples, diacritics stripped). † After runaway correction.

Dataset	Type	Samples	OCR CER
Experiment 1 datasets (line-level)			
PATS-Akhbar	Typewritten	415	2.12%
PATS-Simplified	Typewritten	415	2.94%
PATS-Traditional	Typewritten	415	3.63%
PATS-Arial	Typewritten	415	3.65%
PATS-Naskh	Typewritten	415	4.10%
PATS-Tahoma	Typewritten	415	5.82%
PATS-Thuluth	Typewritten	415	10.21%
PATS-Andalus	Typewritten	415	12.12%
KHATT-val	Handwritten	186	34.24%
Experiment 2 additional datasets (page-level)			
KHATT-Paragraph	HW paragraph	449	61.68%
Yarmouk-testing	Print (encycl.)	1,663	49.85%
Muharaf-val	HW document	116	129.39%†
Historical	HW manuscript	40	105.10%†

III. Datasets and Experimental Setup

A. Datasets

The three experiments use complementary Arabic OCR datasets spanning line-level and page-level documents across typewritten, handwritten, and historical domains. **PATS-A01** [3] is a typewritten dataset of eight font families; we use an aligned 85/15 train/validation split (seed=42), yielding 415 validation samples per font from a shared 2,766-line pool, so that font is the sole cross-font variable. Training splits feed self-reflective analysis (Phase 4); validation splits feed all reported metrics. **KHATT** [4] is a handwritten dataset (34.24% baseline CER), a substantially harder challenge than any typewritten font. Four **extended full-page datasets** (Experiment 2) test page-level generalisation: **KHATT-Paragraph** (449 handwritten paragraph images), **Yarmouk** (1,663 contemporary printed encyclopedic pages), **Muharaf** (116 handwritten documents), and **Historical** (40 degraded archival manuscript pages). Table I summarizes all evaluation datasets.

Diacritics are stripped before CER and WER evaluation to avoid penalizing the model for diacritic insertion or deletion, which is not the target of this correction task. Both diacritics-stripped and raw metrics are computed; we report the stripped variant throughout.[1]

B. OCR Engine and Correction Model

We use Qaari as the OCR engine; its output serves as both the Phase 1 baseline and the input to all LLM correction phases. Qwen3-4B-Instruct-2507 [6], a 4-billion-parameter instruction-tuned model with native Arabic support, is the primary correction LLM. Inference ran on a GPU-accelerated environment (Kaggle T4) with the `enable_thinking` flag set to `False` to suppress the chain-of-thought scratchpad, which adds token cost without measurable benefit here.

C. Evaluation Metrics

Character Error Rate (CER) is the primary metric:

$$\text{CER} = \frac{S + D + I}{N}, \tag{1}$$

where S, D, I are character-level substitutions, deletions, and insertions, and N is the reference character count. Word Error Rate (WER) is reported alongside CER. Lower values indicate better performance. Each phase follows a three-stage pipeline—*export* (local), *infer* (cloud GPU with HuggingFace checkpointing), and *analyze* (local against ground truth)—decoupling correction from evaluation.

IV. Experimental Methodology

The study is organized into the phases of Table II. Phase 1 characterizes the OCR errors and builds the per-dataset confusion matrix and morphological error taxonomy; Phase 2 is the zero-shot LLM correction baseline. Phases 3, 4, 5, 8, and 9 each introduce exactly one knowledge source under an *isolation design*—changing one variable relative to Phase 2, so any phase is beneficial only if it beats Phase 2's CER. Phase 6 tests combinations of the isolated strategies with ablation. All correction prompts use a conservative, XML-structured system template ("return as-is if correct; fix only clear visual OCR errors; do not add diacritics"). The RAG phases (8, 9) retrieve the top-$k = 5$ most similar (OCR, GT) pairs from the Phase 2 *training-split* corrections—a same-engine, same-font, same-domain corpus—and inject them as a per-sample few-shot block; Phase 8 retrieves by BM25 [17] over character n-grams ($n \in \{2, 3, 4\}$), Phase 9 by error-signature similarity (CAMeL morphological profile + confusion profile). Preliminary external-corpus retrieval (OpenITI [18]) degraded severely from domain mismatch, motivating the same-domain corpus.

[1] Results with diacritics retained show the same relative patterns.

TABLE II
EXPERIMENTAL PHASES. ALL CORRECTION PHASES USE THE CONSERVATIVE ZERO-SHOT PROMPT AS THEIR BASE. PHASE 5 RESULTS ARE EXCLUDED FROM COMPARISONS (IMPLEMENTATION BUGS; CORRECTED RE-IMPLEMENTATION EVALUATED IN EXPERIMENT 2).

Ph.	Method	Knowledge source	Novel element
1	OCR baseline + taxonomy	Qaari output, CAMeL	Confusion matrix, valid-but-wrong rate
2	Zero-shot correction	none (conservative prompt)	Comparison baseline
3	OCR-aware prompting	confusion matrix (P1)	Failure-filtered confusions
4	Self-reflective	P2 training-split errors	Self-diagnostic stats + word pairs
5	CAMeL post-proc.	morphological analyzer	*excluded—implementation bugs*
6	Combinations	P3+P4 (+P5)	Ablation of components
8	RAG (BM25)	training corrections	Char n-gram retrieval
9	Error-sig. RAG	CAMeL + confusion + training	Error-structural retrieval

V. Results

A. OCR Baseline and Zero-Shot Correction

Table III reports the Qaari OCR baseline and the Phase 2 zero-shot correction for all nine Experiment 1 datasets. The OCR error rate varies $5.7\times$ across PATS fonts (2.12–12.12% CER, macro-average 5.57%), so typeface choice has a larger effect than any post-processing we test; KHATT (34.24%) is $6.1\times$ harder. Morphological analysis shows 71% of OCR errors are valid-but-wrong (real but contextually incorrect Arabic words) and 29% are non-words, bounding what a morphological revert can catch. Zero-shot correction is net harmful on PATS (macro-average 5.57% $\rightarrow$ 5.84%, +4.8%), degrading seven of eight fonts; only Traditional benefits (-0.67 pp), and since Traditional (3.63%) is helped while Arial (3.65%) is harmed, a simple CER threshold cannot explain the pattern—error type and systematicity matter. On KHATT, zero-shot gives a consistent but modest -0.99 pp, as segmentation failures resist text-only correction.

B. Phases 3–9: Isolated Knowledge Augmentation

Table IV compares all isolated strategies against the Phase 2 baseline, macro-averaging across PATS-A01 fonts with KHATT reported separately. Phase 3 (confusion matrix) and Phase 4 (self-reflective) give small PATS gains (-0.10 and -0.25 pp vs. P2); Phase 4 is the strongest isolated PATS strategy, reaching near-parity with the OCR baseline (5.59% vs. 5.57%), and is uniquely able to improve the lowest-error font by explicitly warning the model against over-correction. Phase 8 (domain-matched BM25 RAG) achieves the best PATS-A01 result (5.45%, -0.39 pp vs. P2, -0.12 pp vs. OCR)—the only strategy to beat the OCR baseline—while Phase 9 (Error-Signature RAG) achieves the best KHATT result (33.13%, -1.11 pp). Phase 5 (CAMeL) is excluded: post-

TABLE III
OCR BASELINE (PHASE 1) AND ZERO-SHOT LLM CORRECTION (PHASE 2), DIACRITICS STRIPPED. ΔCER IS P2−OCR IN PERCENTAGE POINTS; NEGATIVE INDICATES IMPROVEMENT. FONTS ORDERED BY INCREASING OCR CER.

Dataset	OCR CER	P2 CER	ΔCER
PATS-Akhbar	2.12	2.71	+0.59
PATS-Simplified	2.94	2.93	-0.01
PATS-Traditional	3.63	2.96	-0.67
PATS-Arial	3.65	4.50	+0.85
PATS-Naskh	4.10	4.27	+0.17
PATS-Tahoma	5.82	6.13	+0.31
PATS-Thuluth	10.21	10.92	+0.71
PATS-Andalus	12.12	12.27	+0.15
PATS-A01 Macro-Avg	5.57	5.84	+0.27
KHATT-val	34.24	33.25	-0.99

TABLE IV
ISOLATED AUGMENTATION STRATEGIES VS. PHASE 2 ZERO-SHOT BASELINE. PATS-A01 CER IS THE UNWEIGHTED MEAN ACROSS ALL EIGHT FONTS; KHATT SEPARATE. DIACRITICS STRIPPED. BEST LLM RESULT PER COLUMN IN **BOLD**. † PHASE 5 EXCLUDED (IMPLEMENTATION BUGS).

	PATS-A01 (avg)		KHATT	
Phase	**CER**	**ΔP2**	**CER**	**ΔP2**
1 (OCR baseline)	5.57	—	34.24	—
2 (Zero-shot)	5.84	+0.27	33.25	-0.99
3 (OCR-Aware)	5.74	-0.10	33.26	-0.98
4 (Self-Reflective)	5.59	-0.25	33.24	-1.00
5 (CAMeL)†	21.89	n/a	41.62	n/a
6 (conf_self)	5.76	-0.08	33.27	-0.97
8 (RAG, BM25)	**5.45**	**-0.39**	33.83	-0.41
9 (Error-Sig RAG)	5.47	-0.37	**33.13**	**-1.11**

hoc analysis found the morphological analyser was silently disabled and the revert keyed by token rather than position, so its values (21.89% PATS, 41.62% KHATT) are invalid.

C. Phase 6: Combinations and Ablation

Table V reports strategy combinations. RAG strategies dominate: Phase 8 is best on PATS (5.45%, the only strategy beating the OCR baseline) and Phase 9 best on KHATT (33.13%). The `conf_self` combination (P3+P4) is weaker on PATS (5.76%) than P4 alone (5.59%), indicating that adding confusion-matrix context to an already self-aware model slightly increases intervention on clean fonts. The Phase 8 ablation against zero-shot shows RAG eliminating harm on the highest-error fonts (Andalus −1.86 pp, Thuluth −1.24 pp) while only marginally worsening Akhbar—the domain-matched corpus suppresses false corrections where they hurt most.

D. Per-Font Analysis

The best per-font strategy (Table VI) confirms that no single approach dominates: `conf_self` wins on the lowest-error fonts, Error-Signature RAG on the mid-range, and BM25 RAG on the highest-error fonts (Thuluth, Andalus). Two regimes emerge—on Akhbar no LLM strategy beats raw OCR (clearest case of the over-correction threshold), whereas on Traditional zero-shot achieves an 83.1% relative reduction, the single largest improvement in the study. The optimal strategy tracks baseline error rate, supporting selective deployment.

TABLE V
PHASE 6: STRATEGY COMBINATIONS VS. PHASE 2 ZERO-SHOT BASELINE. CONF=CONFUSION-MATRIX INJECTION, SELF=SELF-REFLECTIVE. PATS-A01 CER IS THE UNWEIGHTED MACRO-AVERAGE. DIACRITICS STRIPPED. BEST PER COLUMN IN **BOLD**.

	PATS-A01 (avg)		KHATT	
Combination	**CER**	**ΔP2**	**CER**	**ΔP2**
P2 (zero-shot)	5.84	—	33.25	—
conf_only (P3)	5.74	-0.10	33.26	-0.01
self_only (P4)	5.59	-0.25	33.24	-0.01
conf_self (P3+P4)	5.76	-0.08	33.27	+0.02
P8 (RAG, BM25)	**5.45**	**-0.39**	33.83	+0.58
P9 (Error-Sig RAG)	5.47	-0.37	**33.13**	**-0.12**

TABLE VI
BEST LLM STRATEGY PER FONT (CER, DIACRITICS STRIPPED). Δ IS VS. OCR BASELINE IN PERCENTAGE POINTS. THE OPTIMAL STRATEGY VARIES ACROSS FONTS, CONFIRMING THAT NO SINGLE APPROACH DOMINATES UNIVERSALLY.

Font	OCR	P2	Best	Best Strategy
Akhbar	2.12	2.71	2.46	P6 conf_self
Simplified	2.94	2.93	2.72	P6 conf_self
Traditional	3.63	2.96	**2.75**	P6 conf_self
Arial	3.65	4.50	3.76	P9 Error-Sig
Naskh	4.10	4.27	3.94	P9 Error-Sig
Tahoma	5.82	6.13	5.76	P9 Error-Sig
Thuluth	10.21	10.92	9.68	P8 RAG
Andalus	12.12	12.27	10.41	P8 RAG
KHATT	34.24	33.25	33.13	P9 Error-Sig

E. Experiment 2: Extended Prompt Design Study (13 Datasets)

Experiment 2 crosses four prompt styles (base, conservative, error-pattern, error-pattern+conservative) with two retrieval modes (zero-shot / BM25 RAG), giving the eight trials in Table VII. **T3 (conservative zero-shot)** achieves the best PATS CER (5.27%, −5.4% relative vs. OCR), beating all eight Experiment 1 phases—including P8 RAG (5.45%)—without retrieval overhead, showing that conservative framing is more effective than any knowledge augmentation on clean typewritten text. **T2 (base+RAG)** is the best overall configuration (5.44% PATS, 33.82% KHATT) and is selected for Experiment 3. Error-pattern trials (T4–T8) are brittle; T7 fails catastrophically (190.60% CER) due to prompt-template tag injection.

TABLE VII
EXPERIMENT 2 TRIAL DEFINITIONS AND PATS-A01 MACRO-AVG / KHATT RESULTS (NORMAL SAMPLES, DIACRITICS STRIPPED). † T7: TEMPLATE INJECTION FAILURE, EXCLUDED. BEST PATS IN **BOLD**.

#	Style	RAG	PATS CER	KHATT CER
OCR	baseline	—	5.57%	34.24%
T1	Base	No	5.84%	33.25%
T2	Base	Yes	5.44%	33.82%
T3	**Cons.**	**No**	**5.27%**	32.93%
T4	Cons.	Yes	27.02%	32.69%
T5	Err-pat.	No	5.85%	32.90%
T6	Err-pat.	Yes	19.64%	32.64%
T7†	Ep+Cons	No	190.60%	51.64%
T8	Ep+Cons	Yes	46.56%	32.54%

TABLE VIII
EXPERIMENT 2 FULL-PAGE DATASET RESULTS: QAARI OCR BASELINE AND T2 (BASE+RAG) CORRECTED CER† AFTER RUNAWAY CORRECTION. QWEN3-4B, NORMAL SAMPLES, DIACRITICS STRIPPED.

Dataset	OCR CER	T2 CER†
KHATT-Paragraph	61.68%	45.06%
Yarmouk-testing	49.85%	43.35%
Muharaf-val	129.39%	89.78%
Historical	105.10%	94.53%
Full-page Avg	**86.50%**	**68.18%**

For the four full-page datasets (Table VIII), T2 with runaway correction (threshold 3.0×GT length) reduces the average CER from 86.50% to 68.18% (−21.2% absolute)—a far larger gain than on line-level PATS or KHATT, driven mainly by the runaway corrector on domain-diverse page-level input.

F. Experiment 3: Multi-Model, Multi-OCR-Source, Multi-Domain

Experiment 3 applies T2 across four LLMs, two OCR sources, and two held-out benchmarks. In the model comparison (Table IX), Gemma-3-12B is best on PATS (5.33%) while Qwen3-4B is best on KHATT (33.82%) and full-page (68.18%); Qwen3-14B suffers a KHATT runaway epidemic (175.69% raw, recovered to 35.55%), so larger is not better within Qwen3.

For the OCR-source comparison (Table X), Gemma-3 VLM OCR is excluded from PATS/KHATT (line-strip images are squashed by its preprocessor, yielding hallucinations >170% CER). On full-page inputs, Qaari and Gemma OCR are comparable on average (86.50% vs. 87.33%) and converge after correction (68.18% vs. 64.28%), so OCR-source choice matters less once the LLM corrector is applied.

Correction generalises robustly to both held-out benchmarks (Table XI): on RDI-Test-Lines, Qaari 95.31% → 85.11% (−10.7% abs) and Gemma 68.49% → 61.13%, with Gemma OCR outperforming Qaari overall—its general VLM capability transfers better to unseen styles. On Kitab, correction

TABLE IX
EXPERIMENT 3A: FOUR MODELS ON VALIDATION SET WITH QAARI OCR AND T2. CER† = RUNAWAY-CORRECTED. FP = FULL-PAGE GROUP AVG.

Model	Params	PATS†	KHATT†	FP†
OCR Baseline	—	5.57%	34.24%	86.50%
Qwen3-4B	4B	5.44%	**33.82%**	**68.18%**
Qwen3-14B	14B	5.60%	35.55%	88.51%
Gemma-3-4B	4B	7.47%	36.14%	71.12%
Gemma-3-12B	12B	**5.33%**	35.98%	70.01%

TABLE X
EXPERIMENT 3B: QAARI VS. GEMMA-3 VLM OCR ON FULL-PAGE DATASETS (QWEN3-4B, T2). CER† = RUNAWAY-CORRECTED. GEMMA EXCLUDED FOR PATS/KHATT (LINE-STRIP PREPROCESSING, INVALID).

Dataset	Qaari OCR		Gemma OCR	
	OCR	Corr†	OCR	Corr†
KHATT-Paragraph	61.68%	45.06%	36.14%	35.82%
Yarmouk-testing	49.85%	43.35%	95.76%	74.37%
Muharaf-val	129.39%	89.78%	141.15%	72.85%
Historical	105.10%	94.53%	76.26%	74.09%
Avg	**86.50%**	**68.18%**	**87.33%**	**64.28%**

reduces CER by −17.3% (Qaari) to −24.4% (Gemma) relative. The over-correction threshold reappears in new domains: low-CER subsets (kitab-arabicocr 1.80%, kitab-patsocr 4.79%) are harmed, while kitab-khattparagraph (205.27%) gains most from runaway correction.

VI. DISCUSSION

A. The Over-Correction Threshold

The font-dependent results point to a quantifiable *over-correction threshold* e^*. Let r_f denote the model's fix rate and r_i its introduction rate (fraction of correct tokens wrongly changed). The expected output CER is $\text{CER}_{\text{out}} \approx e(1-r_f) + (1-e)\,r_i$, so correction is net beneficial when $e{\cdot}r_f > (1-e){\cdot}r_i$, yielding $e^* = r_i/(r_f + r_i)$. Empirically $r_i \approx 0.22$, $r_f \approx 0.76$ on PATS-A01, giving $e^* \approx 22\%$ in aggregate; the per-font crossing between Akhbar (harmed) and Simplified (helped) places the effective threshold near 6–7% CER, reflecting a drop in r_f at low error densities. Crucially, adding context (confusion lists, diagnostics) signals to the model that the input is defective and raises r_i, pushing e^* up and making correction harmful over a wider range; conversely Phase 4 explicitly lowers r_i, which is why it is the only strategy to improve the lowest-error font and why `conf_self` synergises (r_f up via P3, r_i down via P4).

B. Practical Implications and Limitations

For clean text (CER <8%), the conservative prompt outperforms every knowledge-augmentation strategy; such output should be corrected conservatively or left unchanged.

TABLE XI
EXPERIMENT 3C: RDI-TEST-LINES AND KITAB BENCHMARKS (QWEN3-4B, T2, RUNAWAY-CORRECTED). RDI LINE SEGMENTATION TASK EXCLUDED (POLYGON-ONLY GT, NO CER COMPUTABLE). KITAB SHOWS SIX REPRESENTATIVE SUBSETS PLUS OVERALL. ↑ = OVER-CORRECTION (LLM CORRECTION HARMFUL).

Subset	Src	OCR CER	Corr†
RDI-Test-Lines (line recognition only)			
LR-Handwritten	Qaari	98.82%	91.55%
LR-Handwritten	Gemma	**62.53%**	**57.71%**
LR-Manuscripts	Qaari	81.94%	79.00%
LR-Manuscripts	Gemma	78.10%	70.36%
LR-Typewritten	Qaari	105.19%	84.79%
LR-Typewritten	Gemma	64.86%	55.31%
Overall	Qaari	95.31%	85.11%
Overall	Gemma	**68.49%**	**61.13%**
Kitab Benchmark (selected subsets)			
kitab-arabicocr	Qaari	1.80%	7.21%↑
kitab-patsocr	Qaari	4.79%	5.98%↑
kitab-hindawi	Qaari	31.46%	24.31%
kitab-historyar	Qaari	63.48%	51.69%
kitab-muharaf	Qaari	76.04%	58.72%
kitab-khattparagraph	Qaari	205.27%	90.60%
Kitab Avg	Qaari	49.28%	**40.73%**
Kitab Avg	Gemma	79.45%	60.07%

Moderate-to-high-error output benefits from correction, motivating selective deployment. RAG is effective only when the retrieval corpus is domain-aligned; both this and the over-correction threshold generalise to the held-out benchmarks. This study is limited to one primary LLM (Qwen3-4B) and one OCR engine; CER/WER are proxies for downstream utility, which may rank strategies differently, and self-reflective insights may not transfer across models or domains.

VII. CONCLUSION

Four findings emerge from our three-experiment study (thirteen datasets, four models, two OCR sources, two held-out benchmarks; normal samples, diacritics stripped).

First, zero-shot LLM correction is net harmful on typewritten Arabic. Phase 2 raises PATS macro-average CER from 5.57% to 5.84% (+4.8%), harming seven of eight fonts. Traditional (3.63%) is helped while Arial (3.65%) is harmed, showing error type and systematicity—not error rate alone—govern the outcome.

Second, domain-matched RAG is the only Experiment 1 strategy to beat the OCR baseline on PATS. Phase 8 (BM25) reaches 5.45% (−0.12 pp vs. OCR) and Phase 9 the best KHATT result (33.13%). Ground-truth-anchored exemplars cut false corrections on clean input; external-corpus RAG degrades severely, confirming domain alignment as a prerequisite.

Third, conservative prompting outperforms knowledge augmentation for typewritten text. Experiment 2's T3 (conservative zero-shot) reaches 5.27% PATS CER, better than all eight Experiment 1 phases including RAG, with no retrieval overhead. Reducing intervention tendency beats supplying more correction targets; error-pattern prompts (T4–T8) are brittle and T7 fails catastrophically.

Fourth, LLM correction generalises robustly to unseen domains. With T2, RDI-Test-Lines improves −10.7% absolute and Kitab −17% to −24% relative. The over-correction finding holds in new domains, so multi-font, multi-domain evaluation is essential—single-dataset conclusions are systematically misleading.

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